Serum concentration and renal excretion by normal adults of inorganic sulfate after acetaminophen, ascorbic acid, or sodium sulfate

Depletion of endogenous inorganic sulfate can have pronounced effects on the elimination kinetics and metabolic fate of phenolic drugs. Our purpose was to determine the effects of acetaminophen (which is partly metabolized to acetaminophen sulfate), ascorbic acid (subject to more limited sulfation than acetaminophen), and sodium sulfate (useful for sulfate repletion by the oral route) on the serum concentration and renal excretion of inorganic sulfate in healthy adults. Six men and two women, 26 to 35 yr old, were studied on four occasions that were at least 4 days apart. They received no medication, 1.5 gm acetaminophen, 6 gm ascorbic acid, or 9 gm sodium sulfate decahydrate orally, in aqueous solution. A blood sample was obtained 2 hr later and urine was collected from 1 to 3 hr. Serum inorganic sulfate concentrations (mean $\pm$ SD), 0.410 $\pm$ 0.043 mM in the control period, were decreased after acetaminophen (0.311 $\pm$ 0.043 mM, $P < 0.001$), increased after sodium sulfate (0.513 $\pm$ 0.055 mM, $P < 0.001$), and apparently unchanged after ascorbic acid (0.417 $\pm$ 0.059 mM). The urinary excretion of inorganic sulfate was decreased after acetaminophen and increased after sodium sulfate. The renal clearance of endogenous creatinine was not affected by any of the treatments. The renal tubular reabsorption of inorganic sulfate is capacity limited, as evidenced by the decrease of the reabsorbed fraction with increasing glomerular filtration rate of the anion ($r = -0.54$, $P < 0.005$). This saturable reabsorption facilitates sulfate homeostasis.

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Man has a limited capacity to metabolize certain drugs to sulfate conjugates. Such limited capacity may, in theory, be a reflection of the saturability of the enzyme systems that activate inorganic sulfate or transfer the activated sulfate to the drug, or it may be due to a limited availability of endogenous inorganic sulfate. Studies on animals have shown pronounced depletion of inorganic sulfate after phenolic drugs that are subject to conjugation with sulfate.[19, 22, 27] This depletion was associated with pronounced changes in the elimination kinetics and metabolic fate of these drugs, changes that could be reversed on repletion of inorganic sulfate by administration of sodium sulfate or of a

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biologic sulfate donor such as cysteine or N-acetylcysteine.[14, 25]

Information concerning the effect of drugs on endogenous inorganic sulfate levels in man is very limited.[15] Our purpose was to determine the effects of acetaminophen (a widely used analgesic that is partly metabolized to acetaminophen sulfate),[26] ascorbic acid (a vitamin often used in very large doses and partly metabolized to ascorbate-2-sulfate),[2] and sodium sulfate (a source of sulfate suitable for oral administration)[9] on the serum concentration and urinary excretion of inorganic sulfate. Since sulfate is eliminated largely by urinary excretion,[3, 42] the investigation was also designed to determine the renal clearance of inorganic sulfate under the various experimental conditions.

Methods

Clinical protocol. Our subjects were six men and two women who were 26 to 35 yr old (mean, 30 yr) and weighed 45.5 to 97.7 kg (mean, 75.3 kg), with an estimated body surface area of 1.41 to 2.24 m^2 (mean, 1.92 m^2) and serum creatinine concentrations of 0.68 to 1.31 mg/dl (mean, 0.93 mg/dl). They were studied on four occasions, at least 4 days apart, and had not taken any drug for at least 72 hr before each study day.

One of the following treatments was administered on different study days, in random order: (a) 100 ml water at 0 and 1 hr, (b) 1.5 gm acetaminophen as Tylenol Extra Strength Adult Liquid (McNeilab) with enough water to make 100 ml at 0 hr, and 100 ml water at 1 hr, (c) ascorbic acid, 3 gm in 100 ml water at 0 and again at 1 hr, for a total dose of 6 gm, and (d) sodium sulfate, 4.5 gm in 100 ml water at 0 and again at 1 hr, for a total dose of 9 gm as the decahydrate (equivalent to a total of 4 gm anhydrous Na_2SO_4). The treatments were administered in the morning, 1 to 2 hr after a light, low-fat breakfast.

The subjects emptied their bladders at 1 hr and urine was collected from the first to the third hour of the study. A 20-ml blood sample was drawn at 2 hr from an antecubital vein by plastic syringe. Saliva was collected at the same time. The blood was permitted to clot and serum was separated by centrifugation. The serum, saliva, and urine samples were immediately frozen and stored at $-20°$ pending assay.

Assay methods. Inorganic sulfate concentrations were determined in triplicate (usual coefficient of variation $\leq 5\%$) by a modification[20] of the turbidimetric assay of Berglund and Sörbo,[7] with a sample volume of 0.3 ml. Creatinine concentrations were determined with a commercial kit (No. 555-A, Sigma), that is based on the modified picrate method of Heinegard and Tiderstrom.[17] Acetaminophen and its conjugates and ascorbic acid had no apparent effect on the creatinine and inorganic sulfate assays; sodium sulfate did not affect the creatinine assay.

Data analysis. Renal clearance of inorganic sulfate and creatinine were determined as excretion rate divided by midpoint serum concentration and were normalized on the basis of body surface area, which was estimated from body weight and height by means of a nomogram.[11] Clearance ratios are renal clearance of inorganic sulfate divided by renal clearance of endogenous creatinine. Since sulfate in plasma or serum is completely ultrafiltrable,[6] the glomerular filtration rate (GFR) of inorganic sulfate was calculated as the product of serum sulfate concentration and renal clearance of creatinine. The renal tubular reabsorption rate of inorganic sulfate was estimated by subtracting urinary excretion rate from filtration rate. The fraction reabsorbed is reabsorption rate divided by filtration rate.

Ascorbic acid study. A separate study was conducted to determine, in greater detail, the effect of ascorbic acid on the urinary excretion of inorganic sulfate. Eight healthy subjects (six men and two women; 25 to 35 years old, some of whom also participated in the studies described in preceding paragraphs), who had been medication-free for at least 1 wk collected their urine every 2 hr from 9 A.M. to 9 P.M. on 2 different days. Their intake of fluids was controlled at 200 ml every 2 hr. On one study day they took a total of 6 gm ascorbic acid in aqueous solution as equally divided doses at 1 and 2 P.M. The other study day served as a control. Five of the subjects repeated the ascorbic acid part of the study once. All participants were permitted to have their regular meals. Inorganic

Table I. *Serum concentration, urinary excretion, and renal clearance of inorganic sulfate after oral acetaminophen, ascorbic acid, or sodium sulfate**

	Inorganic sulfate		*Renal clearance (ml/min/1.73m²)*	
Treatment	*Serum concentration (mM)*	*Urinary excretion (mmol/1.73m²)*†	*Inorganic sulfate*	*Endogenous creatinine*
None (control)	0.410 ± 0.043	1.55 ± 0.46	31.7 ± 10.1	122 ± 18
Acetaminophen (1.5 gm)	0.311 ± 0.043 (P < 0.001)	1.02 ± 0.48 (P < 0.025)	28.0 ± 9.2 (NS)	114 ± 20 (NS)
Ascorbic acid (6 gm)	0.417 ± 0.059 (NS)	1.60 ± 0.55 (NS)	31.5 ± 8.8 (NS)	127 ± 21 (NS)
Sodium sulfate (9 gm) (as the decahydrate)	0.513 ± 0.055 (P < 0.001)	2.36 ± 0.87 (P < 0.005)	38.4 ± 12.7 (NS; P < 0.05‡)	126 ± 27 (NS)

*Results are reported as mean ± SD, n = 8. The statistical significance of differences between treatment and control values is shown in parentheses.

†Amount excreted in the 2-hr period from 1 to 3 hr after drug ingestion.

‡Significance of difference between acetaminophen and sodium sulfate treatments.

sulfate in the urine was determined by the colorimetric method of Haakinen and Haakinen.[16]

Statistical analyses. The experimental data were subjected to analysis of variance,[40] using the BMDP-79 computer program.[12] When significant treatment differences were found, the means were evaluated by the Student–Newman-Keuls test.[40] Regression analysis was performed by the weighted perpendicular least squares method,[36] which is applicable when both variables are subject to error.

Results

The effects of acetaminophen, ascorbic acid, and sodium sulfate on the serum concentration, urinary excretion, and renal clearance of inorganic sulfate are summarized in Table I. Acetaminophen caused a reduction of the serum concentration and urinary excretion rate of endogenous inorganic sulfate, ascorbic acid had no apparent effect, and sodium sulfate ingestion induced increased serum concentrations and urinary excretion of inorganic sulfate. None of the treatments had any apparent effect on the renal clearance of endogenous creatinine and, except for occasional loose stools (not diarrhea), there were no adverse effects after sodium sulfate.

Elevation of serum inorganic sulfate concentrations after sodium sulfate administration was associated with an increased renal clearance of the anion (Table I). The apparent decrease of sulfate renal clearance during acetaminophen-induced hyposulfatemia is not significant, but individual renal clearance–serum concentration profiles (Fig. 1) show concentration dependence of sulfate renal clearance. The profile of one individual, a 30-yr-old man, exhibited a much more pronounced serum sulfate concentration dependence than that of the other seven subjects in the study. His serum creatinine concentrations were consistently in the upper range of normal (between 1.28 and 1.31 mg/dl on 4 different study days over about 1 mo) and his endogenous creatinine renal clearance was consistently below normal (between 73.6 and 87.8 ml/min/1.73 m²). The serum creatinine concentrations of the other subjects were consistently under 1.1 mg/dl in six subjects and between 0.87 and 1.19 mg/dl in the seventh subject. All had endogenous creatinine renal clearances above 98 ml/min/1.73 m² at all times.

Renal tubular reabsorption rate–serum concentration profiles of inorganic sulfate for the three subjects in Fig. 1 are shown in Fig. 2. The one unusual subject had a low, concentration-independent reabsorption rate profile, while the other subjects showed increased reabsorption rate with increasing serum sulfate concentration. The reabsorbed fraction of the filtered load decreased with increasing GFR of inorganic sulfate (Fig. 3).

The concentrations of inorganic sulfate in the saliva were negligible (<0.1 mM) under the con-

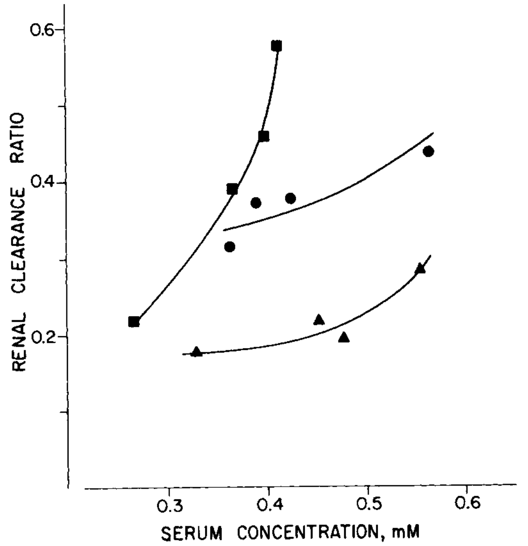

Fig. 1. Relationship between renal clearance ratio (relative to endogenous creatinine clearance) and serum concentration of inorganic sulfate in three subjects. Two of the profiles (▲, female; ●, male) are typical of the group studied while one (■) exhibited a particularly pronounced concentration dependence. The latter profile is from a 30-yr-old man in apparent good health but with a creatinine clearance of only 74 to 88 ml/min/1.73m^2.

ditions of this investigation and are not reported.

A separate study in eight subjects of the effect of a 6-gm oral dose of ascorbic acid on the urinary excretion of inorganic sulfate (involving urine collections every 2 hr over 12 hr) confirmed the absence of any apparent effect of the vitamin on the urinary excretion rate of endogenous inorganic sulfate (results not reported).

Discussion

The mean baseline serum concentration, urinary excretion rate, and renal clearance of inorganic sulfate observed in this study are in good agreement with reported values.[4, 7, 9, 13, 21, 30, 42] Serum concentrations and urinary excretion rates of endogenous inorganic sulfate may be subject to circadian variation[21, 33]; the experiments were therefore scheduled so that blood sampling was always performed in the morning, between 11:00 and 11:30 A.M., to control this variable.

Our results demonstrate that oral administration of a single therapeutic dose of the widely used nonprescription analgesic and antipyretic drug acetaminophen can cause a decrease in the serum concentration and urinary excretion rate of inorganic (free) sulfate in normal young adults. The experimental measurements were made between 1 and 3 hr after drug dosing, i.e., at the time when, according to our preliminary experiments,[24] the sulfate-depleting effect of acetaminophen is at its maximum.

A urinary excretion study had shown that, contrary to prevailing opinion, oral sodium sulfate is reasonably well absorbed.[9] This has now been confirmed by direct determination of serum sulfate concentrations. The increased serum sulfate concentration was associated with an increase in the renal clearance of this anion. The renal clearance–serum concentration profile of inorganic sulfate (Fig. 1) is consistent with capacity-limited renal tubular reabsorption of sulfate. This is evidenced more directly by the negative correlation of the reabsorbed fraction with the GFR of inorganic sulfate ($r = -0.536$, $P < 0.005$; Fig. 3). This concentration dependence, which facilitates sulfate homeostasis, is more definitively demonstrated in rats[29] because it is possible to administer relatively larger doses of acetaminophen to animals and thereby induce more pronounced depletion of endogenous inorganic sulfate.

Ascorbic acid was administered to the subjects in divided doses to increase the bioavailability of this vitamin, which is absorbed by a specialized process of limited capacity.[41] It has been estimated[32] and demonstrated[23] that about one-third of the 6-gm total dose given is absorbed after oral dosing. Ascorbic acid had no apparent effect on serum concentrations and urinary excretion of inorganic sulfate under the conditions of this investigation. To confirm these findings and to rule out the possibility of delayed sulfate depletion due to slow absorption of the vitamin, the experiment was repeated with urine being collected for longer periods of time. Again, there was no apparent effect on the urinary excretion rate of inorganic sulfate. The most likely reason for this lack of effect is that only a small fraction of absorbed ascorbic acid is converted to the sulfate conjugate. One group of investigators has estimated that only about

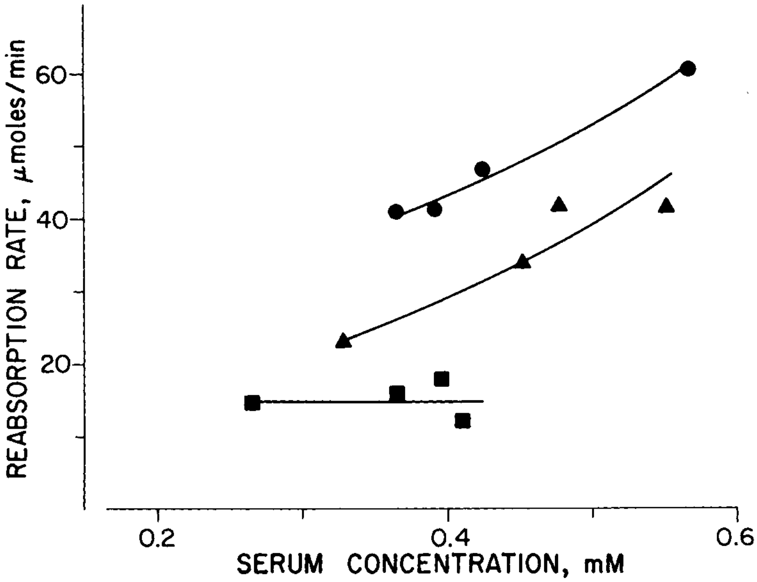

Fig. 2. Relationship between renal tubular reabsorption rate and serum concentration of inorganic sulfate in the three subjects for whom renal clearance ratio–serum concentration profiles are shown in Fig. 1. Symbols are as in Fig. 1.

3% of the normal body pool of the vitamin in humans is metabolized to ascorbate-2-sulfate[31]; others have reported recovery in the urine of 83% of a 1-gm dose of intravenous injected ascorbic acid as such and as metabolites other than the sulfate conjugate and oxalic acid[43]; the fate of the rest of the dose is unknown. Thus, the extent of metabolic sulfation of ascorbic acid may be too limited to cause systemic endogenous sulfate depletion. The results of our investigation do not, however, rule out the possibility of localized depletion (for example, in intestinal tissues[1, 5, 18]) of inorganic sulfate after ascorbic acid and consequent inhibition of drug conjugation with sulfate at that site.

Rapid intravenous infusion of sodium sulfate has revealed the presence of a transport maximum for the renal tubular reabsorption of sulfate in man.[4] One of the subjects in our study exhibited an unusually low transport maximum (Fig. 2) and, consequently, a very pronounced concentration dependence of sulfate renal clearance (Fig. 1). It is not known if this case is rare or if many individuals exhibit similar characteristics. Since the one unusual subject in our study also exhibited subnormal creatinine clearance, the decreased renal tubular reabsorption rate maximum for sulfate may be associated with renal dysfunction.

The results of our investigation demonstrate that even a single therapeutic dose of the widely used nonprescription analgesic and antipyretic drug acetaminophen causes partial, transient depletion of endogenous inorganic sulfate. Greiling and Schuler[15] have shown that oral salicylamide, 50 mg/kg, a drug that also undergoes conjugation with sulfate, causes a reduction of serum inorganic sulfate concentrations in man. Büch et al.[8] found that a 2-gm oral dose of acetaminophen reduced the urinary excretion of inorganic sulfate and increased the excretion of conjugated sulfate (mainly as acetaminophen sulfate), but did not measure serum concentrations and renal clearance of inorganic sulfate. However, Lin and Levy[29] have shown in rats that acetaminophen has no direct effect on the renal clearance of sulfate.

The kinetic implications of endogenous sulfate depletion are well illustrated by acetaminophen. The drug is eliminated largely by conversion to sulfate and glucuronide conjugates, and, to a minor extent, by urinary excretion and conversion to several other metabolites, including one that is responsible for the hepatotoxicity

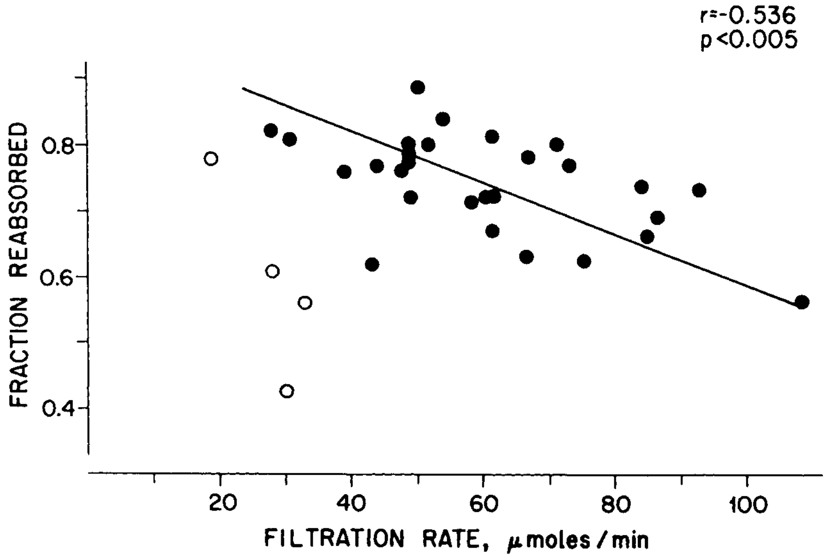

Fig. 3. Relationship between fraction reabsorbed (reabsorption rate/filtration rate) and GFR of inorganic sulfate. Data from eight subjects studied on four occasions. The *open symbols* are data from one unusual subject (■ in Figs. 1 and 2) that were not included in the regression analysis.

of acetaminophen overdose.[10, 34] When large amounts of the drug are taken, the fraction of the dose excreted as acetaminophen sulfate decreases with increasing dose, while the fraction excreted as the terminal metabolite of the hepatotoxic intermediate metabolite increases with increasing dose.[10, 39] At the same time, total clearance of the drug decreases with increasing dose.[35, 39]

Definitive studies in rats, which also eliminate acetaminophen largely by formation of sulfate and glucuronide conjugates, have provided a comprehensive picture of the nonlinear aspects of acetaminophen disposition which, for ethical reasons, cannot be obtained in man.[14] Briefly, as doses are increased from 15 to 300 mg/kg, the fractional conversion of the drug to acetaminophen sulfate decreases and the time-averaged acetaminophen clearance also decreases.[14] Plasma acetaminophen concentrations decline, apparently exponentially, albeit with a decreasing apparent t½ with increasing dose rather than in a manner consistent with Michaelis-Menten kinetics.[14] This is also the case in man.[35] Serum concentrations of inorganic sulfate in rats gradually decrease after acetaminophen, reaching near-zero levels after a 150- or 300-mg/kg dose of acetaminophen.[27] Concomitant administration of sodium sulfate or of a biologic precursor of sulfate prevents this depletion of inorganic sulfate, increases the formation of acetaminophen sulfate, and thereby increases the total clearance of acetaminophen.[14, 27] The plasma concentration–time profile of acetaminophen after 150 or 300 mg/kg together with sodium sulfate has the typical characteristics of Michaelis-Menten kinetics.[14] Thus, the rate-limiting factor in the disposition of acetaminophen during concomitant administration of sulfate is no longer the availability of endogenous cosubstrate (activated sulfate), but the capacity of the sulfate conjugation process (phenol sulfotransferase).

Perhaps even more relevant clinically are the events observed on prolonged infusion of acetaminophen in rats at a rate which, in theory (i.e., based on the kinetics of a small bolus dose), should yield steady-state plasma concentrations similar to the concentration induced immediately after rapid injection of only 15 or 30 mg/kg, (a therapeutic dose in man). Such infusions do not induce a steady-state concentration, but rather cause very substantial drug cumulation to concentrations several times those theoretically predicted based on the clearance of the small bolus dose.[14] Associated with this large drug cumulation is a substantial decrease of endogenous inorganic sulfate concentrations and of the acetaminophen sulfate formation rate. This time-dependent cumulation of acetaminophen and decrease of acetaminophen sulfate formation can be prevented by concomitant use of sodium sulfate or of a biologic precursor of inorganic sulfate.[14] Systemic retention of inorganic sulfate due to impaired renal function reduces the likelihood of acetaminophen-induced sulfate depletion and prevents or decreases acetaminophen cumulation secondary to sulfate depletion.[28]

The preceding description of the dose- and time-dependent kinetics of acetaminophen illustrates the importance of endogenous inorganic sulfate in the disposition of a drug that is subject to conjugation with sulfate. The implied toxicologic importance of physiologic sulfate availability has, in fact, been demonstrated directly; sodium sulfate administration reduced the lethality of single-dose acetaminophen in mice.[38] There are additional toxicologic and physiologic implications: depletion of endogenous inorganic sulfate by a drug such as acetaminophen or salicylamide may affect the disposition of other drugs that are also eliminated by sulfate conjugation[26] (this may include drugs given in much smaller doses that therefore do not ordinarily exhibit capacity-limited elimination kinetics) as well as the disposition of certain endogenous substances that are eliminated partly by sulfation, including steroids, biogenic amines, bile salts, and bilirubin.

Endogenous inorganic sulfate is derived from dietary sources, primarily sulfur-containing amino acids.[22, 37] It is distributed in extracellular space and is eliminated almost entirely by renal excretion.[3, 42] Normal levels of inorganic sulfate in serum are about 1 mM in rats and 0.3 to 0.4 mM in man.[4, 7, 13, 21, 29] The levels in man are lower than those of 13 species of animals tested; no species with serum sulfate levels lower than that of man have been reported.[21] Low sulfate levels in man appear to be due to relatively slow sulfate formation rate rather than to high renal clearance of this anion.[29] These characteristics should make man particularly subject to sulfate depletion by drugs and other xenobiotics that are metabolized to sulfate conjugates. Consideration should be given to the possibility that long-term, heavy use of acetaminophen by individuals on protein-poor diets may lead to sulfate depletion and its potential kinetic and toxic consequences.

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